CS 455: Computer Networks

Last updated Jan 15, 2026

Spring 2026
Prof. Fan Yang

Computer Science, Boston University
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TIME & PLACE TR 9:30-10:45am. 100 Cummington Mall SOC B63

COURSE DESCRIPTION The course introduces concepts underlying the design of high-performance computer networks and scalable protocols. Topics include Internet design principles and methodology, TCP/IP implementation, transport layer security, HTTPS, HTTP/2, HTTP/3, packet switching and performance evaluation. The course uses HTTPS as the main application. Real industry products (such as Google search, CDN, YouTube) will be used as examples to reinforce the concepts and demonstrate various protocols.

LEARNING OUTCOMES In this course, students learn concepts, principles, and protocols used in computer networks, with the Internet as a case study. They learn how to design and implement protocols at many levels of the network architecture and across different timescales. Topics include: distributed inter-process communication; performance measurements; multiplexing; error and flow control; routing; etc. They understand protocol correctness and performance evaluation via statistically reliable measurements and discrete-event simulations. They also become familiar with the Wireshark tool, a network traffic sniffer, and use it to deeply understand the operation of many Internet protocols (DNS, HTTP, TLS, TCP, QUIC, UDP, NAT, IP, Ethernet, ARP, etc.)

PREREQUISITES The course is lectured to undergraduate students.

1. Solid programming skills in a high-level language (such as C, Java, Python) are required. During the lectures I will mostly use C or C-like code, which is more suitable for exposing low-level system details and achieving higher performance in real implementations. Also, if we provide a skeleton of code in some language as part of an assignment, you will be expected to incorporate your code in that language to implement specific networking functions. In this case, the code is mostly straightforward and does not need knowledge of advanced features of the language.
2. A rudimentary understanding of algorithms and their mathematical foundations (CS 112, CS 131) is required.
3. A rudimentary understanding of computer architecture and operating systems (CS 210) is required.
4. A basic understanding of queuing and probabilistic models, discrete-event simulation and how to collect statistically reliable performance metrics (CS 350) is also helpful.

You are expected to already have the background to read and understand code, write and debug reasonably large (1000-line) programs, and learn new syntax and apply it without much difficulty. You are also expected to learn new tools/programs and run them to test and analyze network protocols. If you are in doubt of your background, please talk to the instructor.

CLASS SCHEDULE (tentative)

Dates T	Dates R	Topics
1/20	1/22	Overview: introduction to Internet, protocol, network edge, network core, performance, protocol layers, service models

1/27		
	1/29	Applications: application programming interface (sockets), client-server, DNS, Web (HTTP, cookies, caches) [***Please note: Lecture on 2/18 to substitute Presents' day holiday on 2/17***]
2/3	2/5	
2/10	2/12	Transport services, protocols: UDP, TCP, basics of reliable communication (stop-and-wait, go-back-n, selective-repeat), flow control (sliding window), end-to-end challenges (TCP reliability and flow control), connection management, adaptive retransmission Congestion control: TCP (AIMD, slow start, fast retransmit and recovery)
	2/19	
2/24	2/26	
3/3		
	3/5	Midterm exam
		[***Spring Recess 3/7 to 3/15***]
3/17	3/19	Security: symmetric key cryptography, public key encryption, digital signatures, authentication, TLS
3/24	3/26	
3/31	4/2	
4/7		
	4/9	Routing: packets vs. virtual circuits, distance-vector routing (loops), link-state routing Internetworking: routers, IP datagrams, fragmentation and reassembly, global addressing, forwarding, address resolution (ARP), IPv6, scalability (classless routing, NAT, hierarchical routing), intra- and inter-domain routing protocols (RIP, OSPF, BGP), policy (path-vector) routing
4/14	4/16	
4/21		
	4/23	LANs: encoding, framing, error detection (parity, CRC), MAC protocols (Ethernet CSMA/CD) Wrap-up: Last lecture on 4/30
4/28	4/30	
Final Exam time and location TBD		

OFFICE HOURS Thursdays 11 AM to 12 PM (first OH 2/5). Location: CDS 801

I will adjust my office hour schedule based on student demand to ensure that everyone has access to the support they need.

The purpose of the office hours of the Instructor and Teaching Fellows is to answer specific questions or clarify specific issues. Office hours are not to be used to fill you in on a class you skipped or to explain entire topics. Please come to class!

TEACHING FELLOW and ASSISTANT

Name: Abhinit Sati
Email: abhinit AT bu DOT edu
Office Hours: Wednesday 12 PM - 2 PM
Location: CDS 8th floor (blue corner)

Name: Yuwei Ye
Email: yeyuwei AT bu DOT edu
Office Hours: Friday 4 PM - 5 PM
Location: CDS 362

Grading (except for the final exam) is done by a number of class graders, under the direct supervision of the Teaching Fellow and the Instructor. If you have an issue with a grade (homework or exam), please contact the Teaching Fellow. Only if the issue is not resolved to your satisfaction, please contact the Instructor. Grades must be appealed within one week of receipt.

REQUIRED TEXTBOOK Computer Networking: A Top Down Approach (8th edition), James F. Kurose and Keith W. Ross, Pearson, 2021. You may want to check the cheaper rental option on sites like Barnes & Noble and Amazon.

ONLINE RESOURCES We will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the TF, and the Instructor. Rather than emailing questions to the teaching staff, you are encouraged to post your questions on Piazza. If you have any problems or feedback for the developers, email team@piazza.com.

Find our class page at: https://piazza.com/bu/spring2026/cs455 (**Access code: cqxmmmcbtij**). All course material will also be accessible through Piazza. You should regularly visit Piazza for online references and up-to-date information regarding readings, assignments, exam-related material, announcements, *etc.* You can use the discussion board for asking questions and seeking clarifications, whether from the Instructor, Teaching Fellow, or classmates.
For grade reporting only, we will make use of gradescope.(Entry Code:**PKK46Z**) Details in section "GUIDELINES ON SUBMISSIONS OF ASSIGNMENTS" below.

GRADING POLICY There will be one midterm exam and one final exam. Both exams are closed-book and closed-note, but you are permitted to **bring one A4 sheet of paper with anything written on it**. Also, there will be 6 written homeworks, 6 hands-on labs, and 3-4 programming assignments. You will have 1-2 weeks to complete each assignment. Final grades will be approximately based on the following policy:

- MIDTERM EXAM (25%)
- ASSIGNMENTS (50%): Written (20%), Labs (10%), Programming (20%)
- FINAL EXAM (25%)

Unless otherwise specified, the grading of programming assignments will be generally based on the following policy:

- Program: works correctly (50%); in-line documentation (10%); design quality (10%)
- Design document: how it works (5%); tradeoffs and extensions (5%)
- Testing document: complete compilation instructions (5%); thoroughness of test cases (5%)
- Submission: design & testing documents (10%)

We will use the CS Linux machines (csa1, csa2 and csa3) to grade your lab/programming assignments. Although you may use your own machine, it is your responsibility to ultimately port your assignment to our CS machines to make sure they are graded correctly.

Each assignment will have a due date. If late submissions are allowed for an assignment, there will be a 10% penalty per day for late submissions. But, no late assignments will be accepted after one week from due date, and the last day to submit any late assignments is Wednesday 4/30. Extensions may be granted only for religious holidays and certified medical reasons.

No incompletes will be given, except for reasons of dire illness shortly before the end of the course, and only if a significant amount of work has been completed (*e.g.,* attending lectures, handing in most assignments, and attending the midterm).

Attendance is critical: The intricate and captivating network system is in a perpetual state of rapid evolution. I will periodically deviate from the textbook's content to discuss the most recent developments and advancements in the field. I will not provide backup lecture notes on certain additional details that I will cover in class. While attendance will not directly impact your grade, I strongly encourage you to attend all lectures and take diligent notes.

ACADEMIC HONESTY *All written and programming assignments can be completed either individually or in groups of size at most 2. Discussion of issues in network systems is encouraged, but representing the work of another group as your own is expressly forbidden. This includes "borrowing", "stealing" or "buying" programs/solutions or parts of them (whether in printed or electronic form) from others. We may use an automated plagiarism checker. Cheating will not be tolerated under any circumstances. Handing in your own work a day or two late will affect your grade far less than turning in a copy of someone else's work on time!*

See the CAS Academic Conduct Code, in particular regarding plagiarism and cheating on exams. A student suspected of violating this code will be reported to the Academic Conduct Committee, and if found culpable, the student will receive a grade of "F" for the course.

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College of Arts and Sciences
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ASSIGNMENTS

Dates are tentative, until the assignment is actually posted.

All assignments are to be completed individually. See Academic Honesty on the syllabus.

Tentative Assignment Schedule

Assignment	Posted Date	Due Date
Lab 1 (Wireshark Intro)	**Fri 1/23**	**Mon 2/2**
HW 1 (Getting Started, Chapter 1)	**Fri 1/23**	**Mon 2/2**
PA 1 (Sockets Programming)	**Fri 2/6**	**Mon 2/23**
Labs 2 and 3 (Wireshark DNS and HTTP)	**Fri 2/6**	**Fri 2/20**
HW 2 (Application Layer, Chapter 2)	**Fri 2/6**	**Fri 2/20**
Labs 4 and 5 (Wireshark, UDP and TCP)	**Fri 2/20**	**Mon 3/2**
HW 3 (Transport layer, Chapter 3) (*NO LATE SUBMISSION WILL BE ACCEPTED*)	**Fri 2/20**	**Mon 3/2**
MIDTERM on 3/6		
SPRING RECESS: 3/7 – 3/15		
PA 2	**Mon 3/16**	**Mon 3/30**
Lab 6 (Wireshark TLS)	**Mon 3/23**	**Fri 4/3**
HW 4 (Transport Layer Security, Chapter 8)	**Mon 3/23**	**Fri 4/3**
Lab 7 (IP)	**Mon 4/6**	**Fri 4/17**
HW 5 (IP and Routing)	**Mon 4/6**	**Fri 4/17**
Lab 8 (Ethernet)	**Fri 4/17**	**Wed 4/29**
HW 6 (Ethernet) (*NO LATE SUBMISSION WILL BE ACCEPTED*)	**Fri 4/17**	**Wed 4/29**
FINAL date TBD		

DISCUSSION SECTIONS

The main purpose of the discussion sections is to give you the chance to discuss and ask questions, in a smaller setting under the TF guidance, about specific concepts covered in the lectures or about the assignments, and to allow you to get started on the hands-on Wireshark[1] labs.

[1] Wireshark is a "sniffer" that allows you to observe network traffic so you can understand better how application and network protocols work by observing their operation live!

As part of the homework assignments, you may be asked to submit short reports or answer questions on hands-on labs / exercises.
Each student attends one of the discussion sections held on Tuesdays. **You are expected to use your own laptop and install Wireshark (and other tools) to do the hands-on labs / exercises.**

Tentative Discussion Schedule

Date	Topic
1/23	**Course Overview and Lab 1 Guidance:** Discussion of syllabus and cooperation rule. Introduction to the Internet, protocols, network core, Wireshark basics (Lab 1: Wireshark Intro).
1/30	**Application Layer and HW 1 Q&A:** Protocol layers, network service models, HW 1 (Getting Started, Chapter 1).
2/6	**Sockets Programming and PA 1 Guidance:** Application programming interface (sockets), client-server communication (PA 1: Sockets Programming). Guideline of PA1 and discussion of grading rubric.
2/13	**DNS, HTTP, and Labs 2-3 Guidance:** Deeper dive into DNS and HTTP (Lab 2: Wireshark DNS and Lab 3: HTTP).
2/20	**Application Layer and HW 2 Q&A:** HTTP, cookies, caching mechanisms, HW 2 (Application Layer, Chapter 2).
2/27	**Transport Layer and Labs 4-5 Guidance:** UDP, TCP basics, reliable communication (Lab 4: UDP and Lab 5: TCP).
3/6	**Transport Layer, HW 3 Q&A, and Midterm Review:** Stop-and-wait, go-back-n, sliding window, HW 3 (Transport Layer, Chapter 3), and Midterm preparation.
3/20	**Security and Lab 6 Guidance, and PA 2 Guidance:** Symmetric/public key cryptography, TLS, and Lab 6 (Wireshark TLS). Guideline of PA2.
3/27	**Security and PA 2, HW 4 Q&A:** Digital signatures, authentication, and HW 4 (Transport Layer Security, Chapter 8). Discussion of PA 2 rubrics.
4/3	**Routing and Lab 7 Guidance:** Distance-vector vs. link-state routing, IP addressing, Lab 7 (IP)
4/10	**Routing, IP, and HW 5 Q&A:** Intra- and inter-domain routing protocols (OSPF, BGP), HW 5 (IP and Routing).
4/17	**LANs and Lab 8 Guidance:** Ethernet encoding, error detection (parity, CRC), Lab 8 (Ethernet).
4/24	**LANs and HW 6 Q&A:** CSMA/CD protocols, framing techniques, and HW 6 (Ethernet).
5/1	**Final Exam Review:** Comprehensive discussion on all topics, addressing any remaining questions.

GUIDELINES ON SUBMISSIONS OF ASSIGNMENTS

Assignments will be available on-line.
You are required to **submit an electronic version** of your homework solutions and your programming/lab documentation/reports. We will use Gradescope to grade your submissions online. You can type and submit your write-up in PDF, or if handwritten, you can scan your write-up and submit a PDF. The timestamp on your electronic submission should indicate submission by the due date/time.
Graded assignments will be available online from Gradescope.
Check for Assignments regularly. Start early!

PROGRAMMING / LAB ASSIGNMENTS

You will be required to **submit an electronic copy of your code using the gsubmit program** (as described below). For supporting documentation of your code and any requested written report, you should submit electronic copies using Gradescope (as described above).
When explicitly specified, you may have the option to work in teams of two.

General Requirements on What to Submit

Unless otherwise specified, the program you submit should work correctly and be documented. You should submit an electronic copy of the following:

1. **Program:** a program listing containing in-line documentation (i.e., comments).
2. **Design document:** a separate file (a page or so) describing the overall program design, a verbal description of "how it works", and design tradeoffs considered and made. Also, describe possible improvements and extensions to your program (and sketch how they might be made).
3. **Testing document:** a separate file describing how to run your program. Specify the steps that must be followed to successfully run your program. Also, describe the tests you ran on your program to convince yourself that it is indeed correct. Also, describe any cases for which your program is known not to work correctly.

If a programming assignment asks you to follow a specific design, then you don't have to describe that design in your documentation, but you must still describe possible improvements and extensions.

HOW TO SUBMIT AN ELECTRONIC COPY USING GSUBMIT FROM YOUR CSA ACCOUNT (*Only plain ASCII files!*)

To access csa1/2/3 login nodes, you will need to activate a CS departmental user account if you don't already have one. If you don't have an account and enrolled in this CS course, you can activate your account at https://www.bu.edu/cs/account/

To submit your assignments, use the gsubmit program from your csa account. Submit only your text and source files (*e.g.*, README file and source .java, .c or .py files), but not compiled files (e.g. .class, .o or .pyc files). Do **not** email your assignments.
gsubmit is an electronic file submission engine, which will submit files or directories of files to the grader so they can be marked.
Every file submitted by a given student for a given assignment should have a unique file name. If a file is submitted with a duplicate name, it will either overwrite the file or generate an error message.
To make it easy for the grader to find the files relating to a specific assignment, all files for each assignment should be stored in a subdirectory called lab1, lab2, hw1, hw2, pa1, pa2, etc. and the entire directory should be submitted.

To submit an assignment:

- Create a subdirectory, say "pa#" where # is the assignment number. This is done using the mkdir command: *e.g.*, **mkdir pa5**
- Copy all files necessary for that assignment into the new subdirectory, using the cp command: *e.g.*, **cp prog1.java pa5**
  Be sure to copy only the files you need to submit into this subdirectory.

- Use **gsubmit** to submit the entire subdirectory: **gsubmit cs455 -cp pa5**
  If submission is successful, a status message will be printed.

To submit a file to an already-submitted subdirectory:

If you only submitted part of the assignment and would like to add another file:

- To submit a file README.txt to subdirectory pa5, type (at the prompt):
  gsubmit cs455 -cp README.txt pa5

To resubmit a file:

- To resubmit a file prog1.java in subdirectory pa5, first un-submit (i.e., remove) the file:
  gsubmit cs455 -rm pa5/prog1.java
- Then resubmit it:
  gsubmit cs455 -cp prog1.java pa5

List all files which you have submitted:

- To list all files that you have submitted, type: **gsubmit cs455 -ls**

Looking at a file which has already been submitted:

- To look at a file that has already been submitted, type: **gsubmit cs455 -cat pa5/prog1.java**
- You can store this in a file foo.java by typing **gsubmit cs455 -cat pa5/prog1.java > foo.java**

Where do submitted files go?
Each student who submits an assignment has a subdirectory created to hold his/her files, in a directory for the specified course. This is called the student's “submission spool directory”.
How can the grader tell when a file has been submitted?
Every gsubmit command is automatically logged in a log file, along with a time stamp.
For further information:
Note - The information in this document is taken from the gsubmit man page.
For further information, type: **man gsubmit**

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